

Marco Freschi

POSTDOCTORAL FELLOW IN MECHANICAL ENGINEERING

MFRESCHI@BERKELEY.EDU | [LINKEDIN](#) | [GOOGLE SCHOLAR](#)

RESEARCH INTERESTS

Functional materials | Bioinspired design | Materials for environmental and social impact

PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow - University of California, Berkeley (USA), 2023 - present

"Bioinspired Design of Sustainable Seeding Methods for Improved Ecological Restoration"

Advisor: Prof. Lining Yao

- Implemented the design of the bio-inspired seeding device, increasing the drilling success rate up to 75%.
- Tested the design at the lab-scale, simulating rain and soil conditions, and real-world conditions, with deployment, drilling, and germination tests in California forests and the Kazakh desert.
- Collaborated with partner US universities and an international Foundation, merging multidisciplinary contributions for the improvement of the seeding device design and testing.
- Developed a community-engaged research proposal in partnership with the West Oakland Environmental Indicators Project (WOEIP), focused on soil remediation technology for deployment in low-income neighborhoods

Research Fellow - Politecnico di Milano (Italy), 2023

"Sustainability strategies in the sector of materials for industrial packaging"

Supervisor: Prof. Giovanni Dotelli - *in collaboration with BCube*

- Analyzed the industrial packaging market to evaluate solutions and strategies for decreasing the environmental impact of the sector

Consulting Analyst - Accenture (Italy), 2017-2018

Technical Project Management Office

- Identified and customized the internal IT Operating Model: organization, processes, stakeholders, and use cases.

EDUCATION AND TRAINING

PhD - Politecnico di Milano (Italy), 2022

Materials Engineering

"Production and characterization of self-lubricating metal matrix composites for sliding electrical contact in aerospace applications"

Supervisor: Prof. Giovanni Dotelli - *in collaboration with Logic S.p.A.*

Visiting period at Oulu University (Finland) - Supervisor: PhD O. Haiko, Prof. J. Kömi

- Designed and developed the composite material via powder metallurgy.
- Characterized the composite material through a multiscale approach, from powder morphology to surface and tribological properties, using spectroscopic, microscopic, and mechanical techniques.
- Identified the best composition of copper and micro- and nano- tungsten disulfide solid lubricant for the application, satisfying the requirements of the collaborating company.

Master's degree - Politecnico di Milano (Italy), 2016

Materials Engineering and Nanotechnology

"Experimental investigation of the air path in timber frame wall assembly"

Internship at Université Savoie Mont Blanc – Chambéry (France), resulting in a peer-reviewed publication

Bachelor's degree - Politecnico di Milano (Italy), 2013

Materials Engineering and Nanotechnology

"Anodization of titanium alloy – effect of the operative conditions on the optical and corrosion properties"

PUBLICATIONS

Submitted and in preparation:

- S. Yi, Y. Hong, F. C. Jaeger, **M. Freschi**, Y. Yang, Z. Zhang, H. Lee, K. Soga, E. Mohler, T. Zhang, T. L. Bauerle, S. Yang, L. Yao "Hygrochastic Seedpods: Engineering Microclimates for Ecological Restoration" Nature Sustainability [submitted May 2026]
- **M. Freschi**, S. Yi, L. Yao "Bridging the lab-to-field gap in reforestation technology: a scalable bioinspired seeding system" [in preparation]
- **M. Freschi**, C. Caelli, C. Moletti, L. Primavesi, G. Dotelli "Environmental sustainability of energy storage devices: the production process of aluminum electrolytic capacitors" Journal of Energy Storage, 2026 [in submission]

Published:

- **M. Freschi**, M. Di Virgilio, O. Haiko, M. Mariani, L. Andena, N. Lecis, J. Kömi, G. Dotelli
"Analysis of ball milling time to produce self-lubricating copper-tungsten disulfide composite: Best trade-off between tribological performance and electrical properties"
Next Materials, 2024, 2, 100083. DOI: [10.1016/j.nxmte.2023.100083](https://doi.org/10.1016/j.nxmte.2023.100083)
- **M. Freschi**, L. Dragoni, M. Mariani, O. Haiko, J. Kömi, N. Lecis, G. Dotelli
"Tuning the Parameters of Cu-WS₂ Composite Production via Powder Metallurgy: evaluation of the effects on tribological properties"
Lubricants, 2023, 11(2), 66. DOI: [10.3390/lubricants11020066](https://doi.org/10.3390/lubricants11020066)
- **M. Freschi**, A. Paniz, E. Cerqueni, G. Colella, G. Dotelli
"The twelve principles of green tribology: studies, research, and case studies-a brief anthology"
Lubricants, 2022, 10, 129. DOI: [10.3390/lubricants10060129](https://doi.org/10.3390/lubricants10060129)
- **M. Freschi**, A. Arrigoni, O. Haiko, L. Andena, J. Kömi, C. Castiglioni, G. Dotelli
"Physico-Mechanical Properties of Metal Matrix Self-Lubricating Composites Reinforced with Traditional and Nanometric Particles" Lubricants, 2022, 10, 35. DOI: [10.3390/lubricants10030035](https://doi.org/10.3390/lubricants10030035)
- **M. Freschi**, M. Di Virgilio, O. Haiko, M. Mariani, L. Andena, N. Lecis, J. Kömi, G. Dotelli
"Investigation of second phase concentration effects on tribological and electrical properties of Cu-WS₂ composites"
Tribology International, 2022, 166(107357). DOI: [10.1016/j.triboint.2021.107357](https://doi.org/10.1016/j.triboint.2021.107357)
- **M. Freschi**, M. Di Virgilio, G. Zanardi, M. Mariani, N. Lecis, G. Dotelli
"Employment of Micro- and Nano- WS₂ Structures to enhance the tribological properties of copper matrix composites"
Lubricants, 2021, 9(5), 53. DOI: [10.3390/lubricants9050053](https://doi.org/10.3390/lubricants9050053)
- C. Moletti, **M. Freschi**, L. Primavesi, G. Dotelli
"Comparative Life Cycle Assessment of aluminum electrolytic capacitors"
3rd PCNS Passive Components Networking Symposium Proceedings, 2021, 84-89. ISBN: [978-80-907447-1-4](https://doi.org/978-80-907447-1-4)
- N. Hurel, **M. Freschi**, M. Pailha, M. Woloszyn
"Innovative use of fluorescein for the air path study within light-weight wall assemblies"
Experimental Thermal and Fluid Science, 2018, 92. DOI: [10.1016/j.expthermflusci.2017.10.007](https://doi.org/10.1016/j.expthermflusci.2017.10.007)

MENTORING

Total students supervised: 42

Undergraduate and graduate, UC Berkeley and Politecnico di Milano [2018-present]

University of California, Berkeley – co-supervisor for

Mechanical Engineering – 4 undergraduate students (student assistant) [2025]

Mechanical Engineering – 4 graduate students (capstone project) [2025-2026]

Electrical Engineering and Computer Sciences – 4 undergraduate students (final project class) [2025]

Politecnico di Milano – co-supervisor for thesis projects in

Materials and Chemical Engineering – 7 master's and 23 bachelor's students [2018-2022]

TEACHING ACTIVITIES

Politecnico di Milano – teaching assistant for the courses

School of Architecture (approximately 70 students per year)

Sustainable Materials for Architecture [2021-2022]

Environmental Sustainability and LCA of Building Materials [2020-2021]

Innovative Materials for Architecture [2019-2021]

Science and Technology of Materials [2018-2019]

School of Design (approximately 50 students per year)

Product Performance in Fashion Design [2018-2022]

SERVICE AND LEADERSHIP

Chancellor's Outstanding Staff Award – Nominated staff member selected as recipient, 2026
(University of California, Berkeley)

Berkeley Postdoc Association – Community and Culture Committee Chair, 2026 – present
(University of California, Berkeley)

Berkeley Postdoc Association – Social Committee Co-chair, 2024 – present
(University of California, Berkeley)

ERC Consolidator Grant 2023 Call Reviewer, 2023

Topical Advisor Panel member – Lubricants MDPI, 2023 – present

Peer Reviewer (22), 2019 – present

Applied Sciences (3), Coatings (1), Journal of Bio- and Tribo-Corrosion (1), Lubricants (5), Machines (1),
Materials Circular Economy (1), Polymers (1), RSC Advances (1), Sustainability (4), Tribology Letters (1),
Wear (3)

Research Fellows Representative, 2021 – 2022

Department of Chemistry, Materials and Chemical Engineering Giulio Natta (Politecnico di Milano)

PhD School Representative, 2018 – 2021
(Politecnico di Milano)

CONFERENCES

Tribology 2022 – Barcelona (Spain), 2022

Oral: *Investigation of experimental production parameters effects on the wear behavior of copper-tungsten disulfide composite*

3rd PCNS–Passive Components Networking Symposium – Milano (Italy), 2021

Organizing Committee Member

AIMAT 2021 – XVI Convegno Nazionale – Cagliari (Italy), 2021

Poster: *Effect of second phase concentration on tribological properties of metal matrix self-lubricating composite materials*

K-Trib 2020: 2nd Korea-Tribology International Symposium – Seoul, 2020

Oral: *Employment of micro- and nano-WS₂ structures to enhance the tribological properties of copper-matrix composites*

Materials in the next decade – Favignana (Italy), 2019

Oral: *How to extend the lifetime of electrical sliding contacts through the material design*

Young Materials and Surface Engineers – Roma (Italy), 2019

Poster: *Analysis of the dependence of tribological performances on the properties of the pristine materials*

FUNDED PROJECTS AND FELLOWSHIPS

Bulat Utemuratov Foundation “Scientific Sponsorship Project” – PI: L. Yao – 2026 (Postdoctoral researcher)

Berkeley Climate Action Proof-of-Concept Program – PI: L. Yao – 2025 (Postdoctoral researcher)

Harvard Faculty of Arts and Sciences Research and Academic Exchange – Recipient: M. Freschi – 2025

Enel Foundation Accelerator Postdoctoral Fellowship – Recipient: M. Freschi – 2023-2025

European Institute of Innovation and Technology “Raw Matters Ambassadors at Schools” – PI: G. Dotelli – 2016-2021 (PhD student)

European Institute of Innovation and Technology “ADMA2 – Practical Training Between Academia and Industry During Doctoral Studies” – PI: G. Dotelli – 2018-2021 (PhD student)

TECHNICAL SKILLS

Fabrication: Powder metallurgy | Ball milling | Sintering | 3D-printing | Laser cutting

Characterization: SEM/EDX | XRD | Raman spectroscopy | Optical contact angle | Thermogravimetric analysis | Confocal laser scanning microscopy | Electrical resistance measurements | Uniaxial tensile, compression, and bending testing | Vickers micro-indentation | Tribometer (ball-on-disk, micro-scratch) | Custom environmental testing (rainfall simulation, soil penetration, germination assessment) | Field testing (field deployment in California and Kazakhstan)

Software: Autodesk Inventor | Origin | ImageJ | SimaPro | One Click LCA

Languages: Italian (native) | English (fluent)

MEDIA AND OUTREACH

1. World Forestry Center Discovery Museum, Portland (OR), April 2026 – [Forest Hope Through Innovation](#)
2. CNN Tech for good, December 2025 – [These self-burying ‘E-seeds’ could be the future of reforestation](#)
3. Corriere della Sera, Italy Oct. 2025 – [I semi che si piantano da soli di Berkeley: così la riforestazione può cambiare marcia](#) (The self-burying seeds from Berkeley: how reforestation could take a leap forward)
4. UC Berkeley, Oct. 2025 – [Engineering Nature: Dr. Marco Freschi’s Work on Self-Burying Seed Carriers](#)
5. Victoria and Albert Museum Dundee, Scotland, May 2025 – [Garden Futures: Designing with Nature](#)